

Project Report

1. Title:

YOLO-Based Real Time Object Detection Analysis

2. Objective:

Implementation of YOLO based object detection and analysis of its detection, confidence threshold, inference time and FPS.

3. Tools Used:

Google Colab

Python

Ultralytics YOLO

Matplotlib

Pandas

4. Methodology:

Input Image -> YOLO Model -> Object Detection -> Confidence Score -> Performance Analysis

5. Experimental Result:

Parameter	Result
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Number of Objects detected	8
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Person	7
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Bus	1
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Average inference time	11.20 ms
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Average FPS	89.31
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6. Confidence Analysis

Threshold	Number of Objects
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0.25	8
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0.40	3
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0.50	2
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0.60	2
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0.70	1
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Observations: Increase in confidence threshold resulted in decreased number of detected objects.

7. Conclusion:

In the experiment, it was shown that YOLO is capable of detecting multiple objects from an image along with the confidence score. The experiment obtained an average inference time of 11.20 ms and 89.31 FPS.